Python Class DS-1

Population : The entire group N

Sample : Part of population n

Question → What is the expected age of a student at the time of their graduation ?

sample size = n

population size = N

expected age =

→ mean / avarage

sample mean , $\bar{x} = \frac{x_1 + x_2 + \dots x_n}{n}$

$= \frac{\sum_{i=1}^{n} x_i}{n}$

Population $\mu = \frac{x_1 + x_2 + \dots x_n}{N}$

Outlier : Any data that is very very different or abnormal compare than others.

	Age	
x_1	24	$x_1 - \bar{x} = 0$
x_2	25	$x_2 - \bar{x} = 1$
x_3	26	$x_3 - \bar{x} = 2$
x_4	23	$x_4 - \bar{x} = -1$
x_5	24	
x_6	22	
	27	
	24	

$x_i - \bar{x}$ = Variation of an individual datapoint

Total variation/ Variance $= \sum_{i=1}^{n} (x - \bar{x})$

$$var(x) = \frac{1}{n} \sum_{i=1}^{n} (x_i - \bar{x})$$

sample variance $SV = var(x) = \frac{1}{n} \sum_{i=1}^{n} (x_i - \bar{x})^2$

Population variance $\sigma^2 = \frac{1}{N} \sum_{i=1}^{N} (x_i - \bar{x})^2$

$s^2 = 4$ year^2

$\Rightarrow s = 2$ year

↓

variation means

→ student will graduate 2 years later or 2 years earlier

→ deviates by $s = 2$ years from expected age $\bar{x} = 24$ years

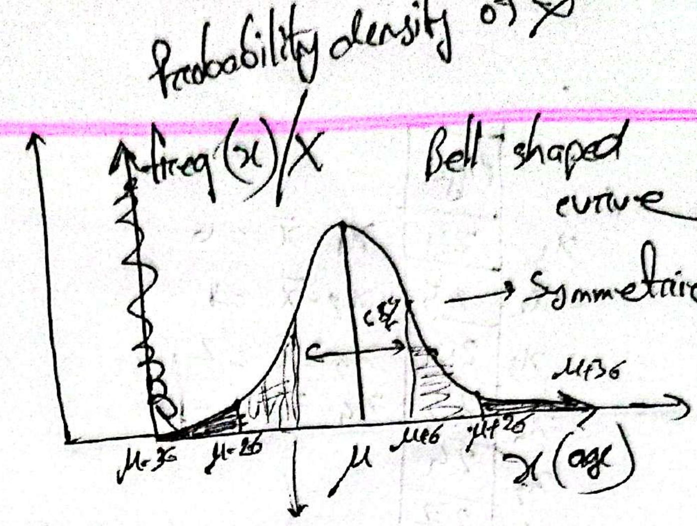
→ Normal rage of ages of studeny $[\bar{x} - s, \bar{x} + s]$

for population variance $[\mu - \sigma, \mu + \sigma]$

Sometime we need to ~~data~~ tolarate more $[\mu - 2\sigma, \mu + 2\sigma]$

standard deviation $\frac{s}{\sigma}$

X	freq(x)



standard deviation $s/\sigma = 2$

Age dataset (x) follows normal distribution.

If any dataset follow Normal distribution
Empirical formula

$X \sim N(\mu, \sigma^2)$ its a function

$\mu = 24, \quad \sigma^2 = 4$

$[\mu - \sigma, \mu + \sigma] \rightarrow 68\%$

$[\mu - 2\sigma, \mu + 2\sigma] \rightarrow 95\%$

$[\mu - 3\sigma, \mu + 3\sigma] \rightarrow 97\%$

Blood pressure average 120

range 120 ± 20

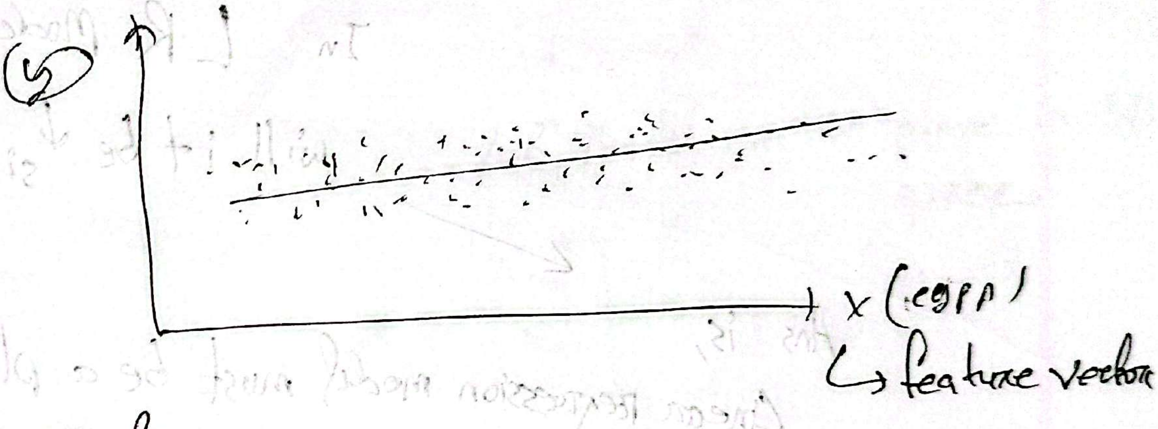
$[100 - 140]$

If we have only one input column/feature column,
↳ simple

DS.1

x (cgpa)	Salary y
x_1	y_1
x_2	y_2
x_3	y_3
⋮	⋮
x_m	y_m

cgpa	salary offered BDT
3.61	75510
3.31	77100



simple linear Regression

we try to fit the best line that will fit

Simple LR Model

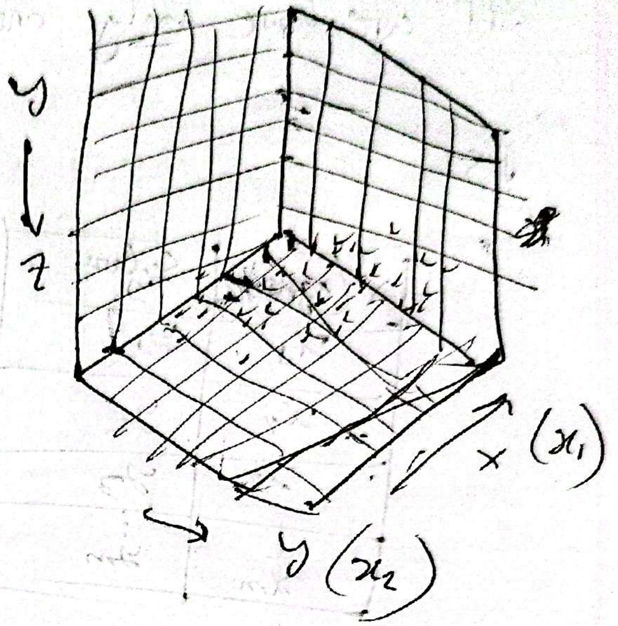
$$\hat{y} = wx + b$$

b → bias/ y intercept

w → slope/coefficient of x

Multiple LR ?

x_1 CGPA	x_2 years of exp	y Salary offered



In LR Model ?

will it be single line ?

Ans is,

Linear regression model must be a plane

If dataset is plotted in 2D space,
then LR will be straight line,

" " " " in 3D space,
then LR will be plane

" " " " " 4D space, LR cube (3D)

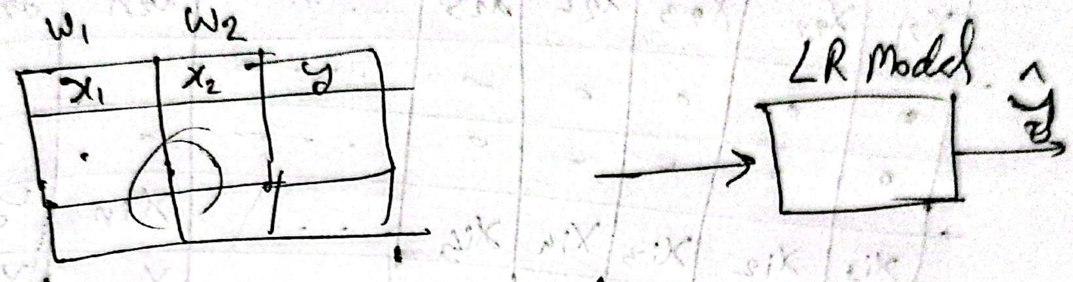
" " " " " 5D , 4D

If LR model is represented → 2D ...

$y = mx + b$

$ax + by + c = 0 \rightarrow$ 2D plane

↳ constant



$\hat{y} = w_1x_1 + w_2x_2 + b$

LR Model = a plane in 3D space

$X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_{2\times1}$, $w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}_{2\times1}$ b = scalar

↓

feature matrix

$w^T = \begin{bmatrix} w_1 & w_2 \end{bmatrix}_{1\times2}$

$\hat{Y} = w^T \cdot x + b$

↓

This is the representation of LR model for any dataset with two features (x_1, x_2) with one target variable y.

$w_1 \quad w_2 \quad w_3$

$x_1, \; x_2, \; x_3, \quad y$

$\hat{y} = w_1x_1 + w_2x_2 + w_3x_3 + b$

$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ $w = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix}$, b

$\hat{y} = w^T X + b$

Multiple LR - 3

ML model Predicted output

Coefficient → w_1, w_2, w_3, w_4, w_5

f_1	f_2	f_3	f_4	f_5	...	f_n	y	$\hat{y}$	
x_{11}	x_{12}	x_{13}	x_{14}	x_{15}	...	x_{1n}	y_1	$\hat{y}_1$	$e_1 = (y_1 - \hat{y}_1)^2$
x_{21}	x_{22}	x_{23}	x_{24}	x_{25}	...	x_{2n}	y_2	$\hat{y}_2$	$e_2 = (y_2 - \hat{y}_2)^2$
.	.	.	.	.	...	.	.	$\hat{y}_3$	$e_3 = (y_3 - \hat{y}_3)^2$
.	.	.	.	.	...	.	.	$\hat{y}_4$	
						x_{in}	y_i	$\hat{y}_i$	
x_{i1}	x_{i2}	x_{i3}	x_{i4}	x_{i5}	...				
					...	x_{mn}	y_m	$\hat{y}_m$	
x_{m1}	x_{m2}	x_{m3}	...						

$$X_i = \begin{bmatrix} x_{i1} \\ x_{i2} \\ x_{i3} \\ \vdots \\ x_{in} \end{bmatrix}$$

→ ith data sample / feature vector

$$X = \begin{bmatrix} x_{11} & x_{12} & x_{13} & \cdots & x_{1n} \\ x_{21} & x_{22} & x_{23} & \cdots & x_{2n} \\ \vdots & \vdots & & & \vdots \\ x_{i1} & x_{i2} & x_{i3} & \cdots & x_{in} \\ x_{m1} & x_{m2} & x_{m3} & \cdots & x_{mn} \end{bmatrix} \quad y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_i \\ y_m \end{bmatrix}$$

$m \times n$

$$x \longrightarrow \boxed{\text{LR Model}} \longrightarrow \hat{y}$$

y_1

$$\hat{y} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_i \end{bmatrix} \begin{matrix} = w_1 x_{11} + w_2 x_{12} + w_3 x_{13} + \dots w_n x_{1n} + b \\ = w_1 x_{21} + w_2 x_{12} + w_3 x_{23} + \dots + w_n x_{2n} + b \\ \\ = w_1 x_{i1} + w_2 x_{i2} + w_3 x_{i3} + \dots w_n x_{in} + b \end{matrix}$$

y intercept

$$W = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ \vdots \\ w_n \end{bmatrix}_{(w \times 1)} \qquad \hat{y}_{(m \times 1)} = X_{(m \times n)} \cdot W_{(n \times 1)} + b$$

$\hat{y}$ → Predicted output Matrix

X → feature Matrix

W → coefficient matrix

$$\boxed{\hat{y} = XW + b}$$

$(X_{(m \times n)}, y_{n \times 1})$

can be the representation of LR model for the given dataset $(X_{m \times n}, y_{m \times 1})$ for the optimal value of b

total error,

$$e = e_1 + e_2 + e_3 + e_4 \dots e_m$$

$$e = \frac{1}{m} \sum_{i=1}^{m} e_i$$

$$\Rightarrow e = \frac{1}{m} \sum_{i=1}^{m} (y_i - \hat{y}_i)^2$$

what will be the value of ~~...~~ $\hat{y}_i$ that will ~~be~~ depend upon w

$$c(\hat{y}_i) \qquad c \to \hat{y}_i \to \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ \vdots \\ w_n \end{bmatrix}$$

$$c(w_1, w_2, \dots \dots w_n)$$

$$c\left(\begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ \vdots \\ w_n \end{bmatrix}\right)$$

c is a function of w matrix

$$c(w) = \frac{1}{m} \sum (y_i - \hat{y}_i)^2$$

↓

target is optimal/minimize the cost function c

Any ~~...~~ LR model and out such value of w and b for which c is minimized

$\underbrace{w \text{ and } b}_{\text{optimal value}}$

↓

need to use Gradient Descent algorithm

$GD(X_{m \times n}, y, \eta)$

1. $w = rand() = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}$
2. $b = rand()$
3. for epoch = 1 to 1000

   $\left.\begin{array}{l} \text{i) compute } \partial e/\partial w \\ \text{ii) compute } \partial e/\partial b \end{array}\right\}$ Batch-GD

   iii) $w = w - \eta * \partial e/\partial w$

   iv) $b = b - \eta * \partial e/\partial b$

return w, b [optimal]

$n \times 1$

$$e = \frac{1}{m}\sum(y_i - \hat{y}_i)^2$$

$$\frac{\partial e}{\partial w} = \frac{\partial e}{\partial \hat{y}_i} \cdot \frac{\partial \hat{y}_i}{\partial w}$$

$$\frac{\partial e}{\partial \hat{y}_i} = \frac{\partial}{\partial \hat{y}_i}\left(\frac{1}{m}\sum_{i=1}^{m}(y_i - \hat{y}_i)^2\right)$$

$$= \frac{1}{m}\sum_{i=1}^{m}\frac{\partial}{\partial \hat{y}_i}(y_i - \hat{y}_i)^2$$

$$= \frac{1}{m}\sum_{i=1}^{m}2(y_i - \hat{y}_i)(-1)$$

$$= -\frac{2}{m}\sum_{i=1}^{m}(y_i - \hat{y}_i)$$

w is a matrix but $\partial e/\partial b$

$$\frac{\partial \hat{y}_i}{\partial w} = \frac{\partial}{\partial w}\left(w_1 x_1 + w_2 x_2 + w_3 x_3 + \dots w_n x_{in} + b\right)$$

y_i = scalar

= (Then class is suddenly stopped)

matrix $\leftarrow \dfrac{\partial \hat{y}_i}{\partial w} \Rightarrow \begin{bmatrix} \frac{\partial \hat{y}_i}{\partial w_1} \\ \frac{\partial \hat{y}_i}{\partial w_2} \\ \vdots \\ \frac{\partial \hat{y}_i}{\partial w_n} \end{bmatrix}$

(n×1)

gradient vector

$$\frac{\partial \hat{y}_i}{\partial w_1} = \frac{\partial}{\partial w_1}\left(w_1 x_{i1} + w_2 x_{i2} + w_3 x_{i3} + \dots w_n x_{in} + b\right)$$

$$= x_{i1} + 0 + 0 + 0$$

↳ because partial derivation

$$\frac{\partial \hat{y}_i}{\partial w_{1}} = x_{i1}$$

$$\frac{\partial \hat{y}_i}{\partial w_2} = x_{i2}$$

$$\therefore \frac{\partial \hat{y}_i}{\partial w} = \begin{bmatrix} x_{i1} \\ x_{i2} \\ x_{i3} \\ \vdots \\ x_{in} \end{bmatrix}$$

$$\underset{(n\times 1)}{\frac{\partial c}{\partial w}} = -\frac{2}{m}\sum_{i=1}^{m}(y_i - \hat{y}_i)\cdot x_i$$

$$= -\frac{2}{m}\begin{bmatrix} (y_1 - \hat{y}_1)x_1 \\ + \\ (y_2 - \hat{y}_2)\cdot x_2 \\ + \\ \vdots \\ + \\ (y_m - \hat{y}_m)\cdot x_m \end{bmatrix}$$

Here $E = \begin{bmatrix} y_1 - \hat{y}_1 \\ y_2 - \hat{y}_2 \\ y_3 - \hat{y}_3 \\ \vdots \\ y_m - \hat{y}_m \end{bmatrix}$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_m \end{bmatrix}_{m \times n} = \begin{bmatrix} x_{11} & x_{12} & x_{13} & \cdots & x_{1n} \\ \vdots & & \cdots & & \cdots \\ x_{m1} & x_{m2} & & & x_{mn} \end{bmatrix}_{m \times n}$$

Formula for Gradient vector

$$\boxed{\underset{(n\times 1)}{\frac{\partial c}{\partial w}} = -\frac{2}{m} \cdot \underset{(n\times m)}{X^T} \cdot \underset{(m\times 1)}{E}}$$

$$= -\frac{2}{m} \cdot \begin{bmatrix} x_1 & x_2 & x_3 & \cdots & x_m \end{bmatrix}^T \begin{bmatrix} y_1 - \hat{y}_1 \\ y_2 - \hat{y}_2 \\ \vdots \\ y_m - \hat{y}_m \end{bmatrix}$$

Now the part for $\frac{\partial c}{\partial b} = \frac{\partial c}{\partial \hat{y}_i} \cdot \frac{\partial \hat{y}_i}{\partial b}$ chain rule of partial differation

$$= -\frac{2}{m} \sum_{i=1}^{m} (y_i - \hat{y}_i) \cdot \frac{\partial}{\partial b} \underbrace{\left(\overset{\text{constant for PD}}{w_1 x_{i1} + w_2 x_{i2} + w_3 x_{i3} + \cdots x_{in} + b} \right)}_{(0+0+0+1)}$$

$$= -\frac{2}{m} \sum_{i=1}^{m} (y_i - \hat{y}_i) \cdot 1$$

$$= -\frac{2}{m} \begin{bmatrix} (y_1 - \hat{y}_1) \cdot 1 \\ + \\ (y_2 - \hat{y}_2) \cdot 1 \\ + \\ (y_i - \hat{y}_i) \cdot 1 \end{bmatrix}_{(m\times 1)} \qquad \text{ones} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ \vdots \end{bmatrix}_{(m\times 1)}$$

$$= -\frac{2}{m} \cdot \text{ones}^T \cdot E$$

$$w = w - \eta * \frac{\partial c}{\partial w}$$

$$\underset{(n \times 1)}{\begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ \vdots \\ w_n \end{bmatrix}} = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ \vdots \\ w_n \end{bmatrix} - \eta * \begin{bmatrix} \partial c/\partial w_1 \\ \partial c/\partial w_2 \\ \vdots \\ \partial c/\partial w_n \end{bmatrix}$$

$$\begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ \vdots \\ w_n \end{bmatrix} = \begin{bmatrix} w_1 - \eta * \partial c/\partial w_1 \\ w_2 - \eta * \partial c/\partial w_2 \\ \vdots \\ w_n - \eta * \partial c/\partial w_n \end{bmatrix}$$

$$w_1 = w_1 - \eta * \frac{\partial c}{\partial w_1}$$

$$w_2 = w_2 - \eta * \frac{\partial c}{\partial w_2}$$

$$\vdots$$

$$w_3 = w_3 - \eta * \frac{\partial c}{\partial w_3}$$

After performing gradient decent algorithm the w and b what we get is optimal

$m \times 1$ $1 \times m$

$X_{(m \times 1)}, y_{m \times 1} \rightarrow \boxed{\text{LR model}} \rightarrow \hat{y} = xw + b$

if w and b are good, then we will get a good $\hat{y}$, thats mean good output

for epoch 1 to 1000:

step 1) $\hat{y}_{m \times 1} = xw + b$

step 2): $E = y_{m \times 1} - \hat{y}_{m \times 1}$

$$ones_{m \times 1} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

step 3) $\frac{\partial c}{\partial w} = -\frac{2}{m} \cdot x^T \cdot E$

$\frac{\partial c}{\partial b} = -\frac{2}{m} \cdot ones^T \cdot E$

step 4) $w = w - \eta * \frac{\partial c}{\partial w}$

$b = b - \eta * \partial c / \partial b$

At one epoch, this 4 steps need to be completed or performed

After 1000 epoch or iteration, what w and b we will get that will be the best optimum value.

for exm, you will be given a dataset of m size,

* teacher may ask to explain the dataset

* Compute the values of w and b running the batch GD on this Dataset up to 2 epoch.

$$X = \begin{bmatrix} 3.61 & 4.5 & 2 & 3 & 27 \\ 3.56 & 4.7 & 3 & 2 & 26 \\ 2.52 & 7.8 & 1 & 4 & 24 \\ 3.76 & 8.0 & 4 & 1 & 30 \\ 2.67 & 7.2 & 2 & 3 & 29 \\ & & 3 & 2 & 28 \end{bmatrix} \quad Y = \begin{bmatrix} 95510 \\ 99100 \\ 60742 \\ 122823 \\ 113068 \\ 117677 \end{bmatrix}$$

(6x5)

$$\begin{array}{cl} w = \begin{bmatrix} 0.12 \\ -0.05 \\ 0.08 \\ 0.03 \\ -0.02 \end{bmatrix} & \begin{array}{l} w_1 \\ w_2 \\ w_3 \\ \\ \end{array} \end{array} \quad b = 0.10$$

these are the random values

η = learning rate $= 10^{-7}$ [as the gradients are large in this problem]

for large → small η

in next class explanation

but it can be anything

Epoch 1

Step 1 : Compute

$$\hat{y} = X\omega + b$$

$$\Rightarrow \hat{Y} = \begin{bmatrix} 0.0182 \\ 0.0722 \\ 0.0324 \\ -0.0748 \\ -0.3096 \\ -0.1156 \end{bmatrix} \begin{matrix} \rightarrow \hat{y}_1 \\ \\ \rightarrow \hat{y}_3 \\ \\ \\ \\ \end{matrix}$$

(chatgpt generating

Step 2 : $E = y - \hat{y} = [\ ] - [\ ]$

$$E = \begin{bmatrix} 93509.9818 \\ 97099.9278 \\ 60741.9676 \\ 122823.0748 \\ 113068.3096 \\ 117077.1156 \end{bmatrix}$$

$$\text{ones} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}_{(6\times 1)} \qquad \text{ones}^T = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

step 3)

$$\partial c/\partial w = -\frac{2}{m} \cdot X^T \cdot E \qquad X^T = \begin{bmatrix} 3.61 & 3.56 \dots \\ 4.5 & 4.7 \dots \\ 2 & 3 \dots \\ 3 & 2 \dots \\ 27 & 26 \dots \end{bmatrix}$$

$$= \begin{array}{c} \begin{bmatrix} -677,626 \\ -12844,226 \\ -539,241 \\ -474,527 \\ -5418,336 \end{bmatrix} \end{array} \begin{array}{l} - \partial c/\partial w_1 \\ - \partial c/\partial w_2 \\ \\ \\ \\ \end{array}$$

$$\partial c/\partial b = -2/m \cdot ones^T \cdot E$$

$$= -2/m \cdot \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

$$\cong 202,773.459$$

step 4)

$$w = w - \eta * \partial c/\partial w$$

$$= \begin{bmatrix} 0.0187762 6 \\ 0.0744926 \\ 0.1339241 \\ 0.0774527 \\ 0.5418336 \end{bmatrix} \begin{array}{l} - w_1 \\ - w_2 \\ - w_3 \\ \\ - w_3 \end{array}$$

$$b = b - \eta \times \partial c/\partial b$$

$$= 0.10 - 0.0202773 \cong 0.0797226$$

Homework → Epoch : 2

After running 1000 or more than ~~10~~ thousand epoch

Corresponding to

$$W \approx \begin{bmatrix} 15489.62 \\ 11353.34 \\ -947.86 \\ 1413.78 \\ -4057.33 \end{bmatrix} \begin{array}{l} w_1 \\ w_2 \\ w_3 \\ w_4 \\ w_5 \end{array}$$

→ it means CGPA will contribute more to predict the output $\hat{y}$, or more important feature, so it is called weight

↓

$W = [\ ]$

weight matrix

w_1 → CGPA → 15489.62

w_2 → years of experience → 1135334

and so on.

$$\hat{y} = XW + b$$

$= w_1 \times CGPA + w_2 \times$ years of exp $+ w_3 \times$ internship count $+ w_4 \times DR + w_5 \times$ age $+ b$

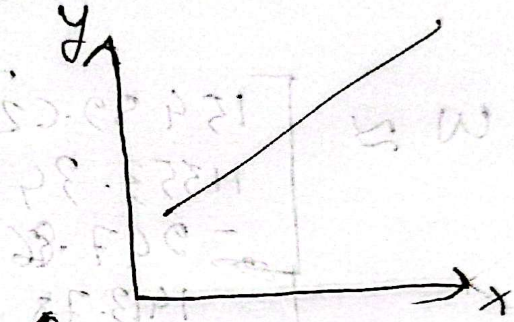
$= 15489 \times CGPA + 11.353.34 \times w_2 \ldots\ldots$

$+ 16[illegible]$

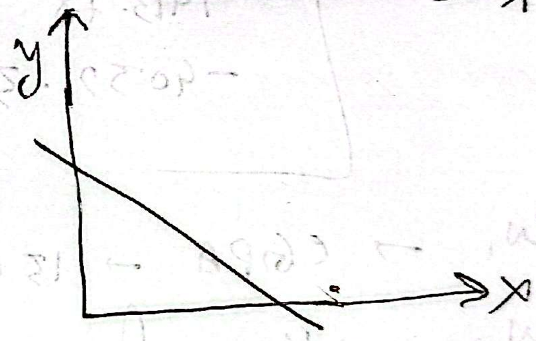
Y intercept is called the bias of the LR model,

and ω is called as weight.

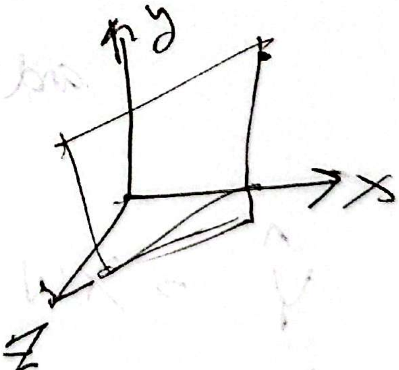
$$y = mx + b$$



$$ax + by + c = 0$$



$$ax + by + cz + d = 0$$



$$\hat{Y} = w_1 x_1 + w_2 x_2 + w_3 x_3 + w_4 x_4 + \dots\ b$$

If we compute ~~z~~ z score of any point x_i that is ~~$~~ not outlier, then the z_i-score ~~must~~ will be between $z \rightarrow [-3, 3]$

Z-score to find out if x_i is an outlier

→ before presenting a dataset to a ML model, we should preprocess the dataset, so that ML model will not predict wrong answer, like removing outliers values.

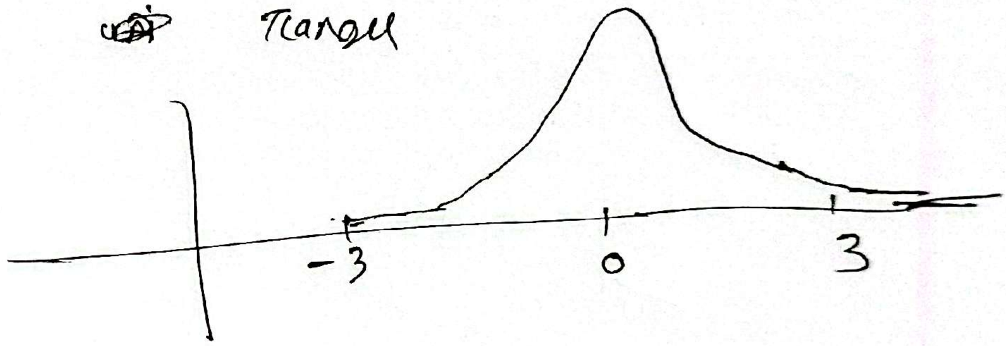
x	z score
24	$\frac{x_i - \mu}{\sigma}$
25	$\frac{x_i - \mu}{6}$
23	
24	
80	

ML course (20	IOT (30)	mark
15	25	

ML 0 ⟷ 20

IoT 0 ⟷ 30

Convert x (age) → z score so that we can compare different data's different graph, in same range

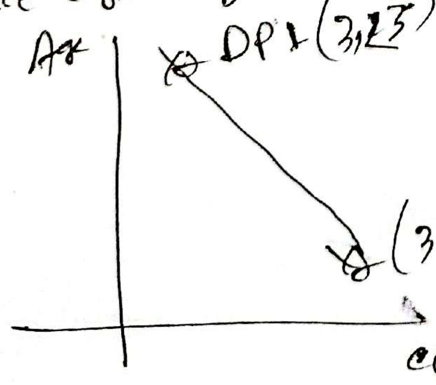


$X \sim N(\mu, \sigma^2)$ → Normal distribution

↓ standarize

$x \sim N(\mu = 0, \sigma^2 = 1)$ → standard of normal distribution

This topic is known as feature engineering



$\sqrt{(3.2-3.6) + (25-50)^2}$

$= \sqrt{(-0.4)^2 + (-25)^2}$

↓

very big

↓

ML will wrong predict

CGPA	AGE	Salary
3.5	40	
3.5	40	
3.6	50	
μ, σ,	μ, σ	→ Solution